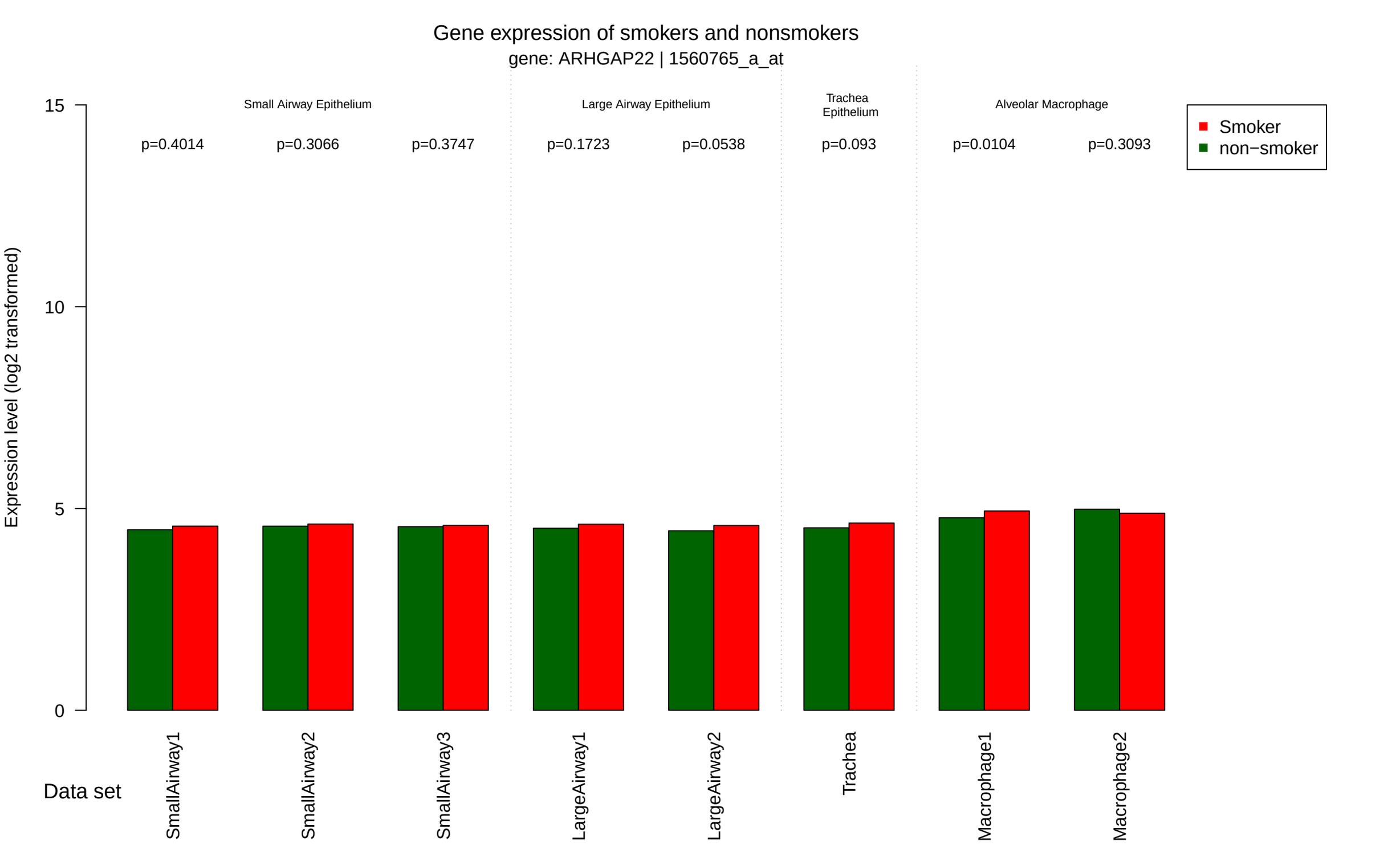
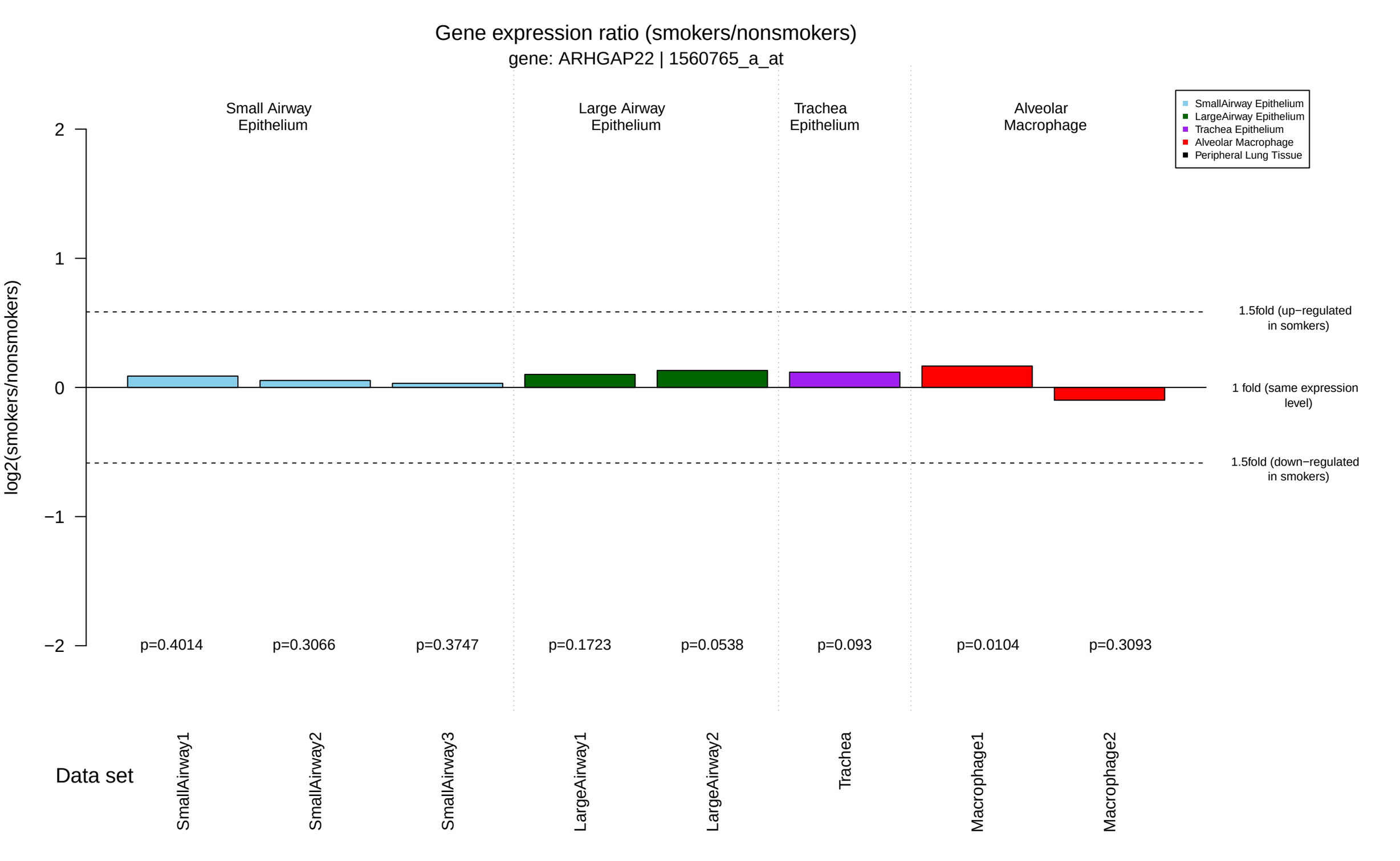
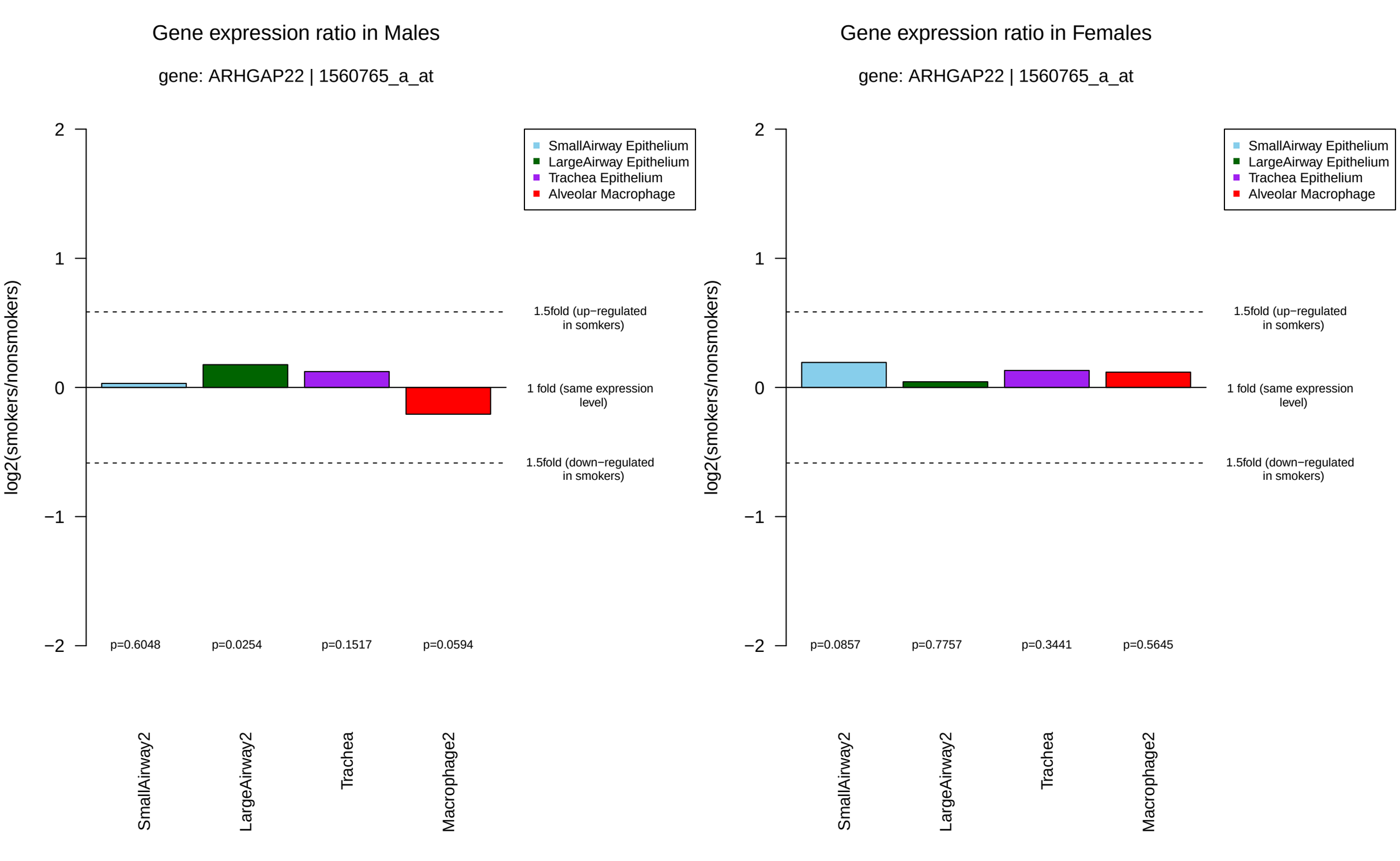
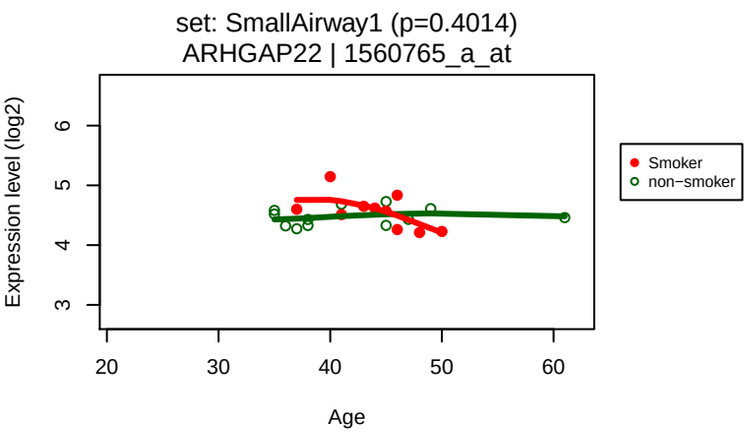
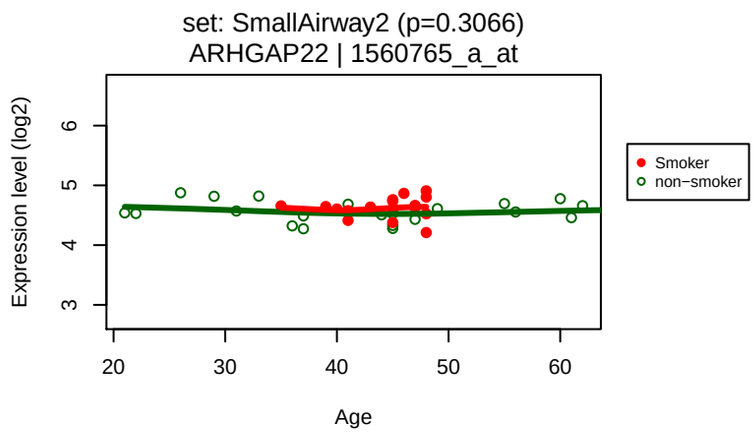
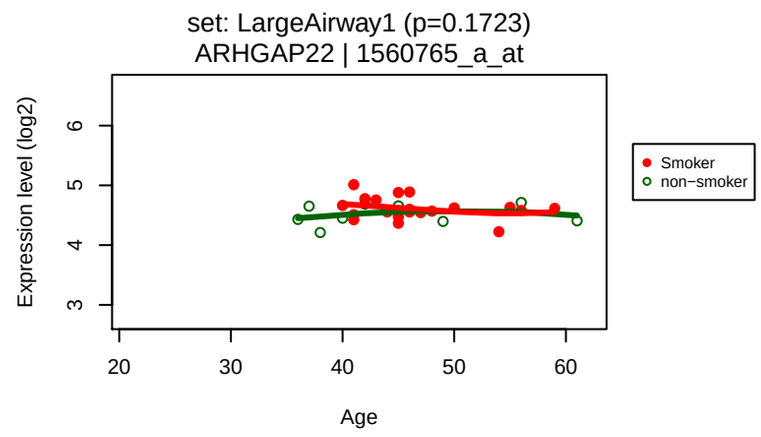

Gene expression of smokers and nonsmokers
gene: ARHGAP22 | 1560765_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4014
p=0.3066
p=0.3747
p=0.1723
p=0.0538
p=0.093
p=0.0104
p=0.3093
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

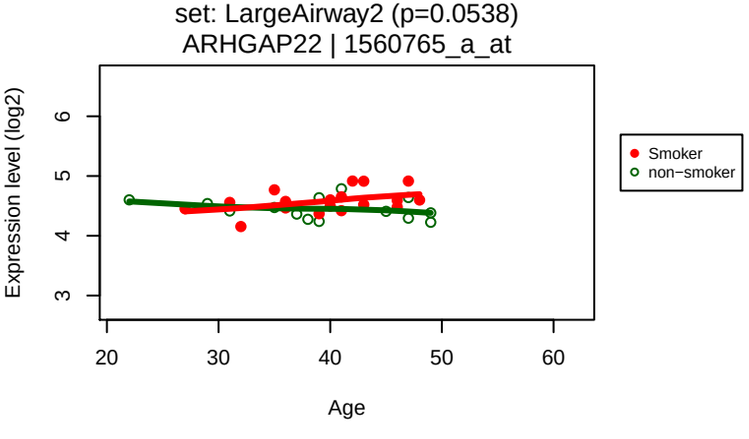

Gene expression ratio (smokers/nonsmokers)
gene: ARHGAP22 | 1560765_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4014
p=0.3066
p=0.3747
p=0.1723
p=0.0538
p=0.093
p=0.0104
p=0.3093
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

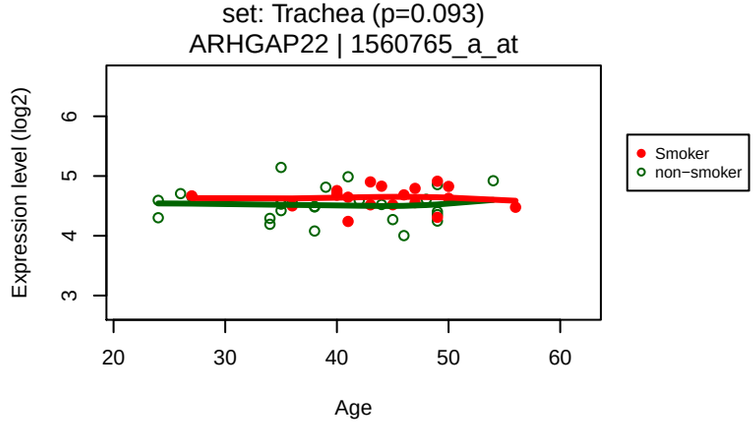

Gene expression ratio in Males
gene: ARHGAP22 | 1560765_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6048
p=0.0254
p=0.1517
p=0.0594
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARHGAP22 | 1560765_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0857
p=0.7757
p=0.3441
p=0.5645
SmallAirway2
LargeAirway2
Trachea
Macrophage2

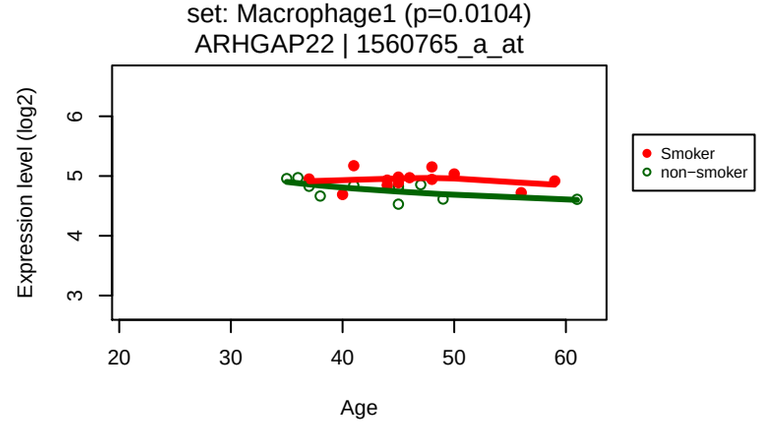

set: SmallAirway1 (p=0.4014)
ARHGAP22 | 1560765_a_at
Expression level (log2)
Age
Smoker
non−smoker

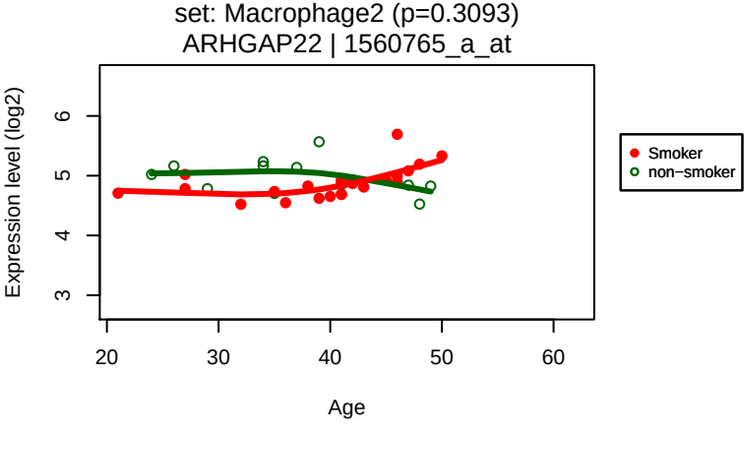

set: SmallAirway2 (p=0.3066)
ARHGAP22 | 1560765_a_at
Expression level (log2)
Age
Smoker
non−smoker

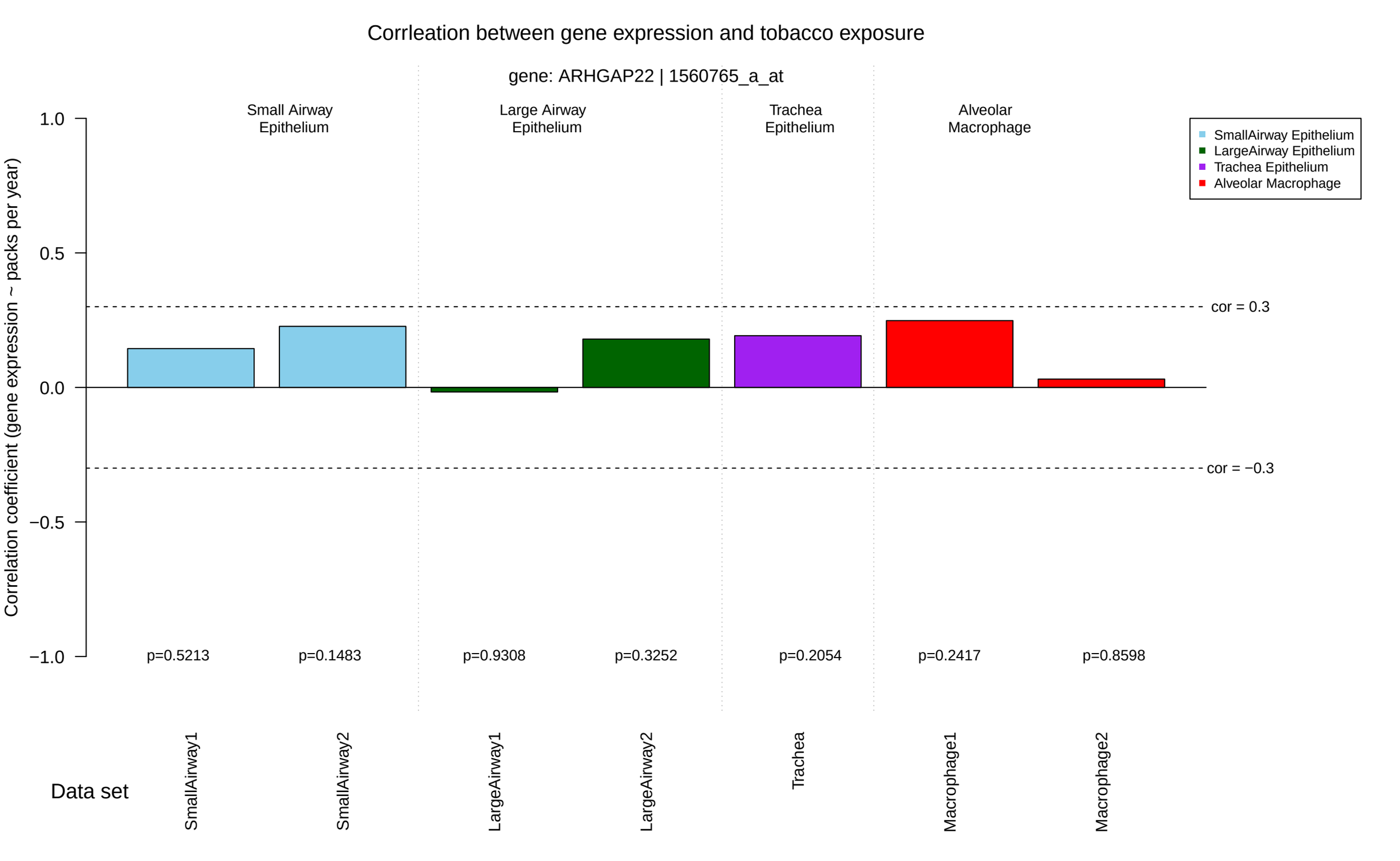

set: LargeAirway1 (p=0.1723)
ARHGAP22 | 1560765_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0538)
ARHGAP22 | 1560765_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.093)
ARHGAP22 | 1560765_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0104)
ARHGAP22 | 1560765_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3093)
ARHGAP22 | 1560765_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARHGAP22 | 1560765_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5213
p=0.1483
p=0.9308
p=0.3252
p=0.2054
p=0.2417
p=0.8598
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2